PDA navigation map

Outlined in blue: the button or scan that leads to another screen. The arrow shows which.

Sign in and start the shift

The person signs in, sees what there is to do and starts. The engine has already chosen the next task.

ROMAFE · PDA-01 · 100% · Smart Warehouse · Welcome · Sign in to start working. · Sign in · Use your username and password. · Username: E.g. 1042 or marta.silva · Password: Enter your password · Sign in · ROMAFE · v1.0.0

01 · Sign in

Identify who is using the PDA, with username and password.

PUT-AWAY · Marta Silva · online · Good morning, Marta · 8 am shift · 4 tasks in your queue · Work · Start tasks · Next: Put away pallet · Operations · Arrival · Fix a location · Look up an item · Look up a location · Queue · Home · Sign out

02 · Put-away home

Show the shift's work and get the person working with one tap; gives access to the look-ups and fixes they do on their own initiative.

Back · Put-away · online · Your queue · 4 tasks · In order. The top one is next. · NEXT Put away pallet · 5W30 oil (40 pcs) · Why: waiting 2h10, the oldest · RECEIVING Receive 2 pallets · Castrol Portugal · Why: they arrived at the dock at 10:05 and are taking it up · CORRECTION Fix 60300302 · Why: 6 pcs of 5W30 oil are in the wrong place · EMPTY PALLET Take the empty pallet to the pallet yard · Why: emptied at 10:02 in 60100201 · Start work

03 · The queue, with the why

Show the tasks in order, and why each one is where it is.

Receive what arrives

The system does not know what is coming: when goods reach the dock, put-away logs the arrival. That creates the receiving task, which the engine puts in the queue by priority. It is checked against the supplier note, goes into stock, and the put-away tasks go back to the engine.

Back · Arrival · online · What arrived? · The system does not know what is coming. Say who brought it and how many pallets. · Scan the note, if it has a barcode · Supplier: Castrol Portugal · PALLETS − 2 + · SKF pallets are still logged on the old PDA. · Log the arrival

04 · Log the arrival

Identify the arrival: the system does not know what is coming, so this is where the goods enter the system.

Back · Arrival · online · Arrival logged at 10:05 · Castrol Portugal · 2 pallets at the dock · SENT TO THE ENGINE · Receiving task: 1 new · Where it is: Inbound dock · The engine puts it in the queue by priority. Until someone starts it, the goods stay at the dock. · See the queue · Back to home

05 · Arrival logged

Confirm the arrival is logged and the receiving task went to the engine.

PUT-AWAY · Marta Silva · online · Receive · Step 1 of 3 · Scan the NOTE · The document that came with the goods, or the supplier's reference · AT THE INBOUND DOCK · 2 pallets · Castrol Portugal · Arrived at 10:05 · Aim at the note and press the trigger · Can't scan — type it

06 · Scan the supplier's delivery note

Start the receiving task: identify the delivery by the supplier's note.

Receive · Step 2 of 3 · Note GR-58213 · Castrol Portugal · Check what arrived · EXPECTED · 5W30 oil · 5 L: 40 pcs · W712 oil filter: 24 pcs · If it matches, confirm. If not, say how many arrived. · It's all here — confirm · The quantity doesn't match

07 · Check what arrived

Compare what was expected with what arrived.

Receive · Step 2 of 3 · 5W30 oil · 5 L · 40 expected · How many arrived? · − 36 + · 4 pieces missing · Recorded as a receiving discrepancy. Receiving goes on. · Only what arrived goes into stock. · Record and continue

08 · The quantity doesn't match

Record how much actually arrived, when it isn't what was expected.

Receive · Step 3 of 3 · Arrival logged at 10:05 · Castrol Portugal · 2 pallets at the dock · SENT TO THE ENGINE · Receiving task: 1 new · Where it is: Inbound dock · The engine puts it in the queue by priority. Until someone starts it, the goods stay at the dock. · See the queue · Back to home

09 · Received

Confirm it is in stock, and that the put-away tasks went to the engine, which sets their priority in the queue.

Put away a pallet

Take the pallet from the engine to the location the engine suggests, and confirm by scanning the item and the location.

Put away pallet · Step 1 of 4 · Scan the ITEM · The label on the box or pallet in front of you · EXPECTED · 5W30 oil · 5 L · 40 pieces · from the inbound dock · Aim and press the trigger · Can't scan — type it

10 · Scan the item

Confirm that the pallet in front of the person is the one in the task.

Put away pallet · Step 2 of 4 · Go to · SUGGESTION 1 OF 3 · 60300301 · Zone 30 · Column 03 · Shelf 01 · WHY HERE · It moves fast — 34 a day — and is 8 m from the dock. Too heavy to go higher. · Route: leave the dock for zone 30, column 03 on the right. · Scan the location · It's taken · Another problem

11 · Go to suggestion 1 of 3

Say where to take the pallet, and why.

Put away pallet · Step 2 of 4 · 60300301 was recorded as taken · The manager will look into why the system thought it was free. · Go to · SUGGESTION 2 OF 3 · 60300401 · WHY HERE · It is the next closest to the dock that takes the weight. · You can only choose another location if the 3rd is taken too. · Scan the location

12 · Number 1 was taken

Record the location as taken and move on to the next suggestion.

Put away pallet · Step 3 of 4 · Scan the LOCATION · The label on the shelf where you placed the pallet · SHOULD BE · 60300401 · When you scan, the system checks again that the location still fits · Can't scan — type it

13 · Scan the location

Confirm, by scanning, that the pallet is in the right location.

Put away pallet · Step 4 of 4 · Put away in 60300401 · 40 × 5W30 oil · recorded under your name at 10:14 · STOCK STATUS · Before: Waiting for put-away · dock · Now: Stored · 60300401 · Next in the queue: Fix 60300302 — stock in the wrong place. · Next task

14 · Put away

Confirm the record and move on to the next task.

When put-away goes wrong

The detours — location taken, doesn't fit, damaged item, label that won't scan. Everything is recorded and the manager sees it.

Put away pallet · Step 2 of 4 · 60300301 · What's wrong? · Location taken · It doesn't fit · Can't reach the location · Damaged item · Unreadable label · Other reason — type it

15 · What's wrong?

Say, with one tap, what is stopping the put away.

Put away pallet · Step 1 of 4 · Type the ITEM · The label won't scan. Type the reference on the box. · Item reference: OL5W30-5 · Recorded as typed in. · Confirm

16 · Type the item

Identify the item when the label won't scan.

Put away pallet · Step 3 of 4 · Type the LOCATION · The eight digits on the shelf label · Location code: 60300401 · Recorded as typed in. If the label keeps failing, the system will advise renewing it. · Confirm

17 · Type the location

Identify the location when the label won't scan, counts towards advising a new label.

Report a problem

Any role reports what is stopping them: it is recorded with who said it and when, and the manager is told.

Put away pallet · Step 2 of 4 · Damaged item · Recorded at 10:16 under your name · The manager has been told. The pallet stays where it is until they decide. · The task leaves your queue. · Next task

18 · Problem recorded

Confirm the problem is recorded and the manager has been told.

Fix a location (task)

The engine found stock in the wrong place and has it moved. It comes out of the source and goes into the destination, in a single record.

Fix a location · Step 1 of 2 · Scan the SOURCE · SHOULD BE · 60300302 · TAKE · 6 × 5W30 oil · 5 L · They are in the wrong place · Scan the location, then the item · It's no longer there

19 · Take from the source

Confirm the location and the item that come out.

Fix a location · Step 2 of 2 · Goes to · 60300401 · Zone 30 · Column 04 · Shelf 01 · WHY HERE · It already holds the same item — it joins what is there. · Scan the location where you put it

20 · Put in the destination

Say where the stock goes, and confirm by scanning.

Fix a location · Step 2 of 2 · 6 × 5W30 oil moved · From 60300302 to 60300401 · at 10:31 · Out and in were a single record. No new stock was created. · Next task

21 · Fixed

Confirm the stock has moved, in a single record.

Fix what is in a location

On their own initiative: whoever sees a location doesn't match the system fixes it right there. Nobody counts.

Back · Fix a location · online · Scan the LOCATION · The label of the location where what's there doesn't match the system · Aim and press the trigger · Can't scan — type it

22 · Scan the wrong location

Identify the location that doesn't match the system.

Back · Fix a location · online · 60300302 · THE SYSTEM SAYS · 5W30 oil · 5 L: 12 pcs · How many are there? · − 6 + · Counts towards the limit of open corrections. · Record the correction

23 · What's there?

Say how many pieces are actually in the location.

Back · Fix a location · online · 60300302 fixed · 5W30 oil: from 12 to 6 pcs · at 10:40 · The manager sees the difference in the shift report. · Back to home

24 · Correction recorded

Confirm the correction; the manager sees the difference.

Take the empty pallet away

The pallet that became empty leaves the location and goes to the yard, the location is freed.

Empty pallet · Step 1 of 2 · Take it to the YARD · The empty pallet from 60100201 · PALLET YARD · Next to the inbound dock · Scan the pallet yard label

25 · Take it to the pallet yard

Take the empty pallet to the yard, and confirm by scanning the label.

Empty pallet · Step 2 of 2 · Empty pallet in the yard · 60100201 became free at 10:45 · Next task

26 · In the pallet yard

Confirm the location is now free.

Look up

Find out where an item is, or what is in a location, without changing anything.

Back · Look up an item · online · Scan the item, or type the reference · 5W30 oil · 5 L · OL5W30-5 · goes to zone 30 · WHERE IT IS · 60300401: 40 pcs · 60300301: 12 pcs · Inbound dock: 36 to put away

27 · Look up an item

See where an item is, and how much is in each place.

Back · Look up a location · online · Scan the location · 60300401 · Zone 30 · Column 04 · Shelf 01 · Item: 5W30 oil · 5 L · Quantity: 40 pcs · Last in: 10:14 · Marta Silva · Isn't this what's there? Fix it here. · Doesn't match — fix it

28 · Look up a location

See what is in a location and, if it doesn't match, fix it.

Starting the picking shift

Picking signs in, sees the round the engine gave them and starts. The batch fills itself and an order goes in whole.

PICKING · Rui Costa · online · Good morning, Rui · 8 am shift · 3 orders in your round · Work · Start picking · Your batch: 3 orders · 18 lines · Operations · Look up an order · Look up an item · Look up a location · Queue · Home · Sign out

29 · Picking home

Show the shift's round and get the person moving with one tap; gives access to the look-ups.

Back · Picking · online · Your queue · 2 tasks · In order. The top one is next. · NEXT Round · 3 orders · 18 lines · Why: the DPD truck closes at 11:30 · EMPTY PALLET Take the empty pallet to the pallet yard · Why: emptied at 9:50 in 60201503 · Start work

30 · Your queue

Show what picking has to do, in the engine's order.

Pick an order

Picking does the round of the warehouse with the batch of orders, and leaves them on the staging shelf.

PICKING · Rui Costa · online · Your batch · 3 orders · 18 lines · 3 columns · DPD TRUCK · CLOSES 11:30 · G-24187 · Auto Peças Norte · 9 lines · tight: 21 min of work left · GLS TRUCK · 14:00 · G-24190 · Oficina Ribeiro · 6 lines · GLS TRUCK · 14:00 · G-24193 · Lubrificantes Sá · 3 lines · Start picking

31 · The batch, already full

Show the orders for the round, already chosen by the engine.

Stop 4 of 18 · G-24187 · Go to · 60200201 · TAKE · 6 × W712 oil filter · For G-24187 · the stock is already reserved for you · Next: 60200502 · 5 m · Scan the LOCATION, then the ITEM · Short quantity · Type it · Another problem

32 · Stop: scan the location

Take the person to the location and say what to pick.

Stop 5 of 18 · G-24190 · Only 2 of the 6 are here · Record what you took. The order stays incomplete and does not ship without the rest. · 60200502 · Ordered: 6 · Taken: 2 · Missing: 4 · It is recorded as a correction to this location. · Record and continue

33 · Short quantity

Record what is missing and fix the stock right there.

End of round · 18 of 18 · Take it to the staging shelf · PP-DPD-03 · For the DPD truck · 11:30 · next to the outbound dock · G-24187: complete · 9 parcels · G-24190 is marked as incomplete · It goes to PP-GLS-01 and stays out of the shipping queue. · Scan the location of the shelf where you place it

34 · Take it to the staging shelf

Leave the orders on the staging shelf, and say where they are.

End of round · 18 of 18 · Round closed at 11:02 · 3 orders on the staging shelf · YOU LEFT · G-24187 · complete: PP-DPD-03 · G-24193 · complete: PP-GLS-01 · G-24190 · incomplete: waiting · Shipping takes the complete ones. The incomplete one waits for stock and does not enter the shipping queue. · Next task

35 · Round closed

Confirm the orders are on the shelf and the round is over.

When picking goes wrong

The detours of the round: nothing in the location, less than the order asks for, a label that will not scan — and leaving the shift with orders still open.

Stop 4 of 18 · G-24187 · 60200201 · What's wrong? · Nothing in the location · Less than the order asks for · A different item is there · Unreadable label · Damaged item · Other reason — type it

36 · What's wrong?

Say, with one tap, what is stopping the pick.

Stop 4 of 18 · G-24187 · Type the LOCATION · The eight digits on the shelf label · Location code: 60200201 · Recorded as typed in. If the label keeps failing, the system will advise renewing it. · Confirm

37 · Type the location

Identify the location when the label won't scan; counts towards advising a new label.

Back · Sign out · online · You have 1 order still open · G-24190 has 4 lines left to pick · IF YOU LEAVE NOW · The order: goes back to the queue · The GLS truck: 2 pm · Whoever comes next picks it up where you left it. The manager sees who left with open work. · Carry on with the round · Leave anyway

38 · Leave with orders still open

Warn that there is unfinished work before the person ends the shift.

Look up (picking)

See how an order is doing, where an item is, or what is in a location, without changing anything.

Back · Look up an order · online · Scan the order or type its number · G-24187 · Auto Peças Norte · DPD truck · 11:30 · Lines: 9 · Picked: 9 · Where it is: PP-DPD-03 · Status: complete

39 · Look up an order

See how an order is doing: what is missing, where it is and which truck it goes on.

Back · Look up an item · online · Scan the item, or type the reference · W712 oil filter · W712 · zone 20 · WHERE IT IS · 60200201: 24 pcs · 60200502: 6 pcs · Reserved for orders: 12 pcs

40 · Look up an item

See where an item is, and how much is in each place.

Back · Look up a location · online · Scan the location · 60200201 · Zone 20 · Column 02 · Shelf 01 · Item: W712 oil filter · Quantity: 24 pcs · Last out: 9:40 · Rui Costa · Isn't this what's there? Fix it here. · Doesn't match — fix it

41 · Look up a location

See what is in a location and, if it doesn't match, fix it.

Starting the shipping shift

Shipping signs in, sees the shift trucks by closing time and starts loading.

SHIPPING · Ana Lopes · online · Good morning, Ana · 8 am shift · 3 camiões hoje · Work · Start loading · Next: DPD truck · closes 11:30 · Operations · Look up an order · See the staging shelf · Queue · Home · Sign out

42 · Shipping home

Show the shift trucks and get the person loading with one tap; gives access to the look-ups.

Back · Shipping · online · Your queue · 2 tasks · By closing time. The top one is next. · NEXT Load the DPD truck · Why: it closes at 11:30 and is already at the dock · CHECK G-24193 na prateleira · Why: it goes on the 2 pm GLS truck · Start work

43 · Your queue

Show what shipping has to do, by truck closing time.

From the shelf to the truck

Shipping checks the parcels, takes them to the dock and loads the truck.

SHIPPING · Ana Lopes · online · Today's trucks · By closing time · DPD · 11:30 · 1 order on the shelf · 0 at the dock · Tight · GLS · 2 pm · 1 complete · 1 incomplete on the shelf · Transportes Vilar · 17:30 · 4 orders to pick

44 · Today's trucks

Show the trucks by closing time, and what is ready.

G-24187 · DPD · Step 2 of 4 · On the shelf: PP-DPD-03 · Scan each PARCEL · Box 1 · filters: V1/9 · Box 2 · oils: V2/9 · Box 3 · oils: V3/9 · Box 4 · brake pads: V4/9 · … 5 more parcels: V5–V9 · 3 of 9 scanned · A parcel is missing · Another problem

45 · Check parcel by parcel

Confirm each parcel before it leaves the shelf.

G-24187 · DPD · Step 3 of 4 · 9 of 9 parcels checked · The order is complete. · Take it to the dock · CAIS-2 · Where the DPD truck is backing in · Scan the label of the dock where you place it

46 · Take it to the dock

Take the parcels to the truck's dock.

DPD dispatch · 11:30 · Step 4 of 4 · Scan while loading · G-24187: 9/9 loaded · G-24181: 4/4 loaded · G-24185: 1/3 · Scan each parcel as it goes onto the truck · Transport documents and long notes: on the shipping computer. · Close dispatch — 2 parcels missing

47 · Load and close the dispatch

Confirm each parcel that is loaded, and close the dispatch.

DPD dispatch · 11:30 · 4 of 4 · DPD dispatch closed at 11:28 · 3 orders · 22 parcels loaded · RECORDED · G-24187: 9 parcels · G-24181: 4 parcels · G-24185: 9 parcels · The transport documents print on the shipping computer. · Next task

48 · Dispatch closed

Confirm the dispatch is closed and what was loaded.

When shipping goes wrong

A parcel missing from the shelf, the truck has not arrived, a label that will not scan — and ending the shift with a dispatch still open.

DPD dispatch · 11:30 · 2 of 4 · PP-DPD-03 · What's wrong? · A parcel is missing from the shelf · Damaged parcel · The truck has not arrived · Unreadable label · It doesn't fit in the truck · Other reason — type it

49 · What's wrong?

Say, with one tap, what is stopping the loading.

DPD dispatch · 11:30 · 2 of 4 · Parcel V4/9 is missing · G-24187 · pastilhas · não está na prateleira · IF IT GOES LIKE THIS · The order: stays incomplete · What goes: 8 of 9 parcels · The rest: waits for the next truck · Picking is told. The manager sees the incomplete order in the shift report. · Record and continue · See the order

50 · A parcel is missing

Record the missing parcel and say what happens to the order.

DPD dispatch · 11:30 · 3 of 4 · Type the DOCK · The number of the dock the truck is backing into · Dock: CAIS-2 · Recorded as typed in. · Confirm

51 · Type the dock

Identify the dock when the label won't scan.

Back · Sign out · online · The DPD dispatch is still open · 2 parcels still to load · IF YOU LEAVE NOW · The dispatch: stays open · The truck: waits at the dock · Whoever comes next picks the dispatch up where you left it. The manager sees who left with open work. · Carry on loading · Leave anyway

52 · Leave with a dispatch still open

Warn that a dispatch is still open before the person ends the shift.

Look up (shipping)

See how an order is doing, and what is on the staging shelf.

Back · Look up an order · online · Scan the order or type its number · G-24187 · Auto Peças Norte · DPD truck · 11:30 · Parcels: 9 · Checked: 8 · Where it is: PP-DPD-03 · Status: incomplete

53 · Look up an order

See how an order is doing: what is missing, where it is and which truck it goes on.

Back · Shelf · online · Scan the shelf location · PP-DPD-03 · Junto ao cais de saída · camião DPD · 11:30 · ON THE SHELF · G-24187: 9 parcels · G-24181: 4 parcels · G-24185: 9 parcels

54 · See the staging shelf

See what is on the shelf and which truck each order goes on.

The manager

The manager sees how the shift is going, reads the signals, assigns tasks with a written reason and sees the whole warehouse.

MANAGER · Paulo Mendes · online · Good morning, Paulo · 8 am shift · 47 tarefas feitas · Work · See the shift · Figures and signals · Operations · Signals · Insights · Assign a task · All operations · Queue · Home · Sign out

55 · Manager home

Show the manager how the shift is going and give access to what only they do.

MANAGER · Paulo Mendes · online · 8 am shift · So far · 10:20 · 47 tasks done · 6% typed instead of scanned · 3 locations taken · 1 incomplete order · SIGNALS · MAP 60300301 refused 3 times · The system thought it was free. Order a fix? · STOCK 60200502 has 2 open corrections · Stock may be leaving unrecorded · LABEL 60300401 · 3 of 10 typed · The label no longer scans well. Renew it?

56 · Shift report

Show the manager the shift figures and the signals that need attention.

Back · Signals · online · TODAY · MAP 60300301 refused 3 times · The system thought it was free. Order a fix? · STOCK 60200502 has 2 open corrections · Stock may be leaving unrecorded · LABEL 60300401 · 3 of 10 typed · The label no longer scans well. Renew it?

57 · Signals

Show what needs attention, and let the manager act with one tap.

Back · Assign · online · TASK · Put away pallet · Óleo 5W30 · It has been in the put-away queue for 2h10 · To whom: Marta Silva · Why: Está na coluna ao lado · The engine records the pattern of assignments and proposes changing the order, but it is management who changes the rule. · Assign

58 · Assign a task

Give a task to a person, with the reason written; the engine learns from it.

Back · Now · online · PUT-AWAY · Marta Silva: putting away 60300401 · In the queue: 4 tasks · At the dock: 2 pallets to receive · PICKING · Rui Costa: stop 4 of 18 · Orders still open: 1 · SHIPPING · Ana Lopes: loading the DPD · Closes at: 11:30 · Assign a task

59 · What is going on

See the whole warehouse: who is working, what is missing and what goes out today.

Back · Done · online · Task assigned to Marta Silva · With the reason written · at 10:22 · She gets it as the next one in the queue. The engine learns from the assignment. · Back to home

60 · Done

Confirm what the manager ordered, and say where it lands.

Back · All operations · online · PUT-AWAY · Log the arrival · Fix a location · Look up an item · Look up a location · PICKING AND SHIPPING · Look up an order · See the staging shelf · The manager does everything the roles do. It is recorded under their name.

61 · All operations

Give the manager everything the roles do, in one place.

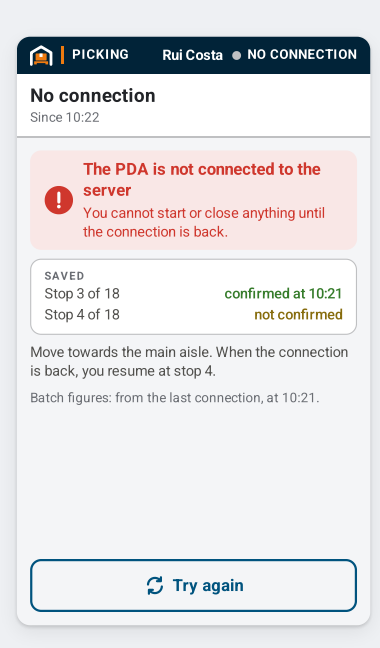
Back · Insights · online · 6% typed instead of scanned · 12 min per put away · LOCATIONS REFUSED, BY ZONE · Zone 30: 7 · Zone 20: 3 · Zone 10: 1 · WHAT THE ENGINE LEARNED · The manager's assignments: 4 this week · Pattern: oils: late in the morning · The engine proposes changing the order of the oils. It is management who changes the rule.

62 · Insights

Show what the data says: where time is lost, what fails and what the engine learned.

When the connection drops

The PDA works connected to Rolgest. With no connection it says it stopped, and shows what was saved.



PICKING

Rui Costa ● NO CONNECTION

No connection

Since 10:22

The PDA is not connected to the server

You cannot start or close anything until the connection is back.

SAVED

Stop 3 of 18 — confirmed at 10:21

Stop 4 of 18 — not confirmed

Move towards the main aisle. When the connection is back, you resume at stop 4.

Batch figures: from the last connection, at 10:21.

63 · No connection

Say the PDA isn't connected, and what was saved.